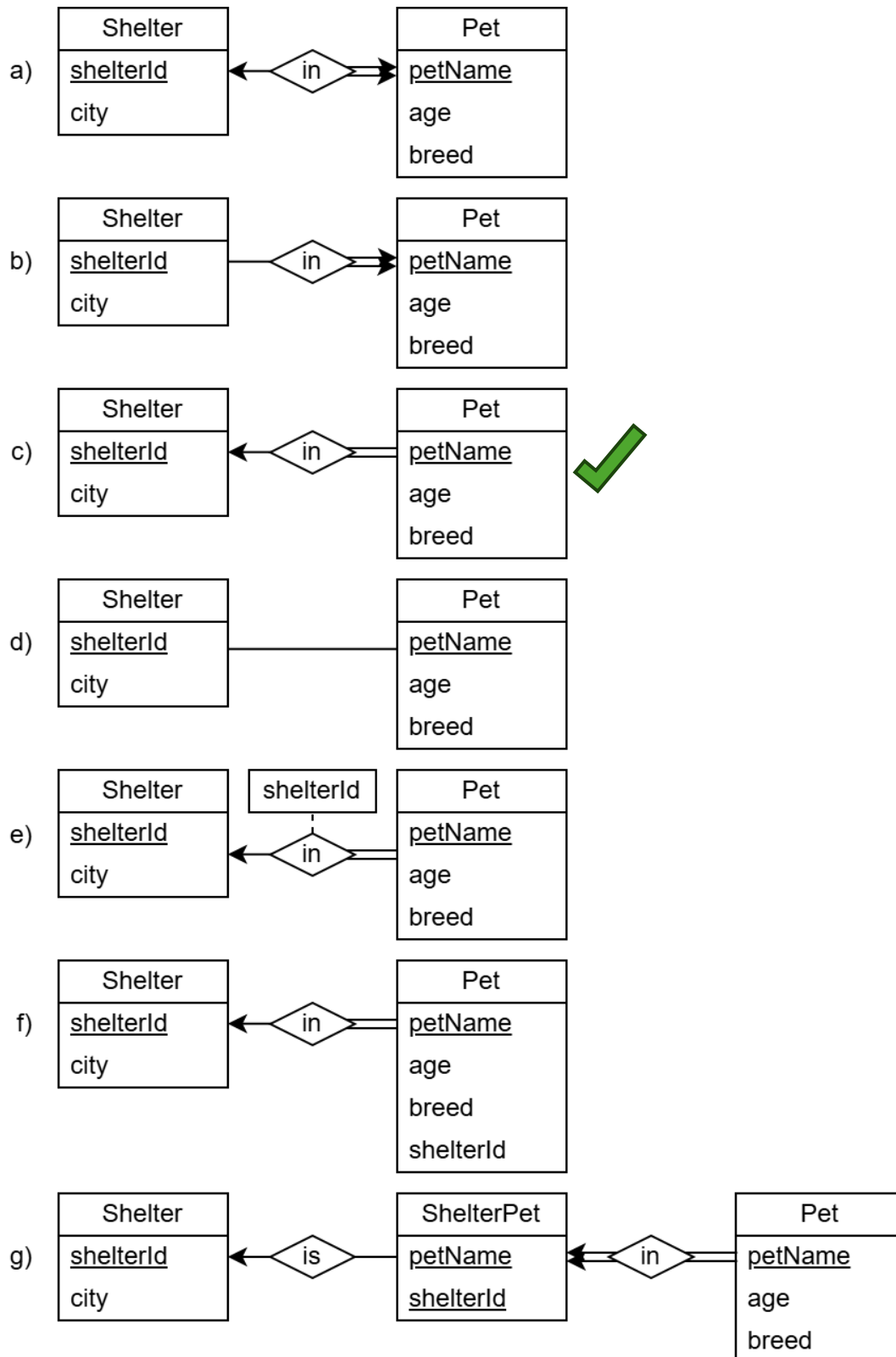


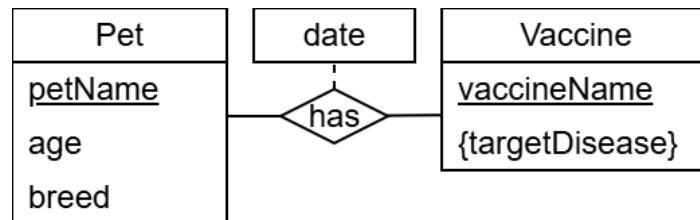
Question 1: ER Modelling

Which of the following ER models correctly depicts the relationship between the entity sets Pet and Shelter?



Question 2: Design

The database is now extended to also save the vaccination status of pets in shelters. Consider the partial ER model below. It implements this change:



Question 2.1: Design Implications

Which of the following statements is true, and which are false?

- a) Every Pet is vaccinated.
- ✓ b) There are vaccines that are not used.
- c) Relationship set *has* forms a ternary association between entity sets *Pet*, *Vaccine*, and *date*.
- d) Every Pet has exactly one vaccine.
- ✓ e) There are vaccines that have been administered to multiple pets.

Question 2.2: Transformation ER to Relational Model

Which of the following options is the correct transformation to the relational model of the above ER model? Foreign keys are not shown and can be ignored for this task.

- a) *Pet*(petName, age, breed)
date(petName, vaccineName, has)
Vaccine(vaccineName, targetDisease)
- b) *Pet*(petName, age, breed)
Vaccine(vaccineName, targetDisease, petName)
- c) *Pet*(petName, age, breed)
has(petName, vaccineName, date)
Vaccine(vaccineName, targetDisease)
- ✓ d) *Pet*(petName, age, breed)
has(petName, vaccineName, date)
Vaccine(vaccineName)
TargetDisease(value, vaccineName)
- e) *Pet*(petName, age, breed, vaccineName)
Vaccine(vaccineName, targetDisease)

Question 3: Normalization

Consider a relation with schema $R(A, B, C, D, E)$, where the primary key is not shown.

The following functional dependencies are assumed:

$$FD = \{A \rightarrow BCD, CD \rightarrow E\}$$

Question 3.1: Primary Key

What is / are the candidate key(s) of R ?

There is only one candidate key: $\{A\}$: A always has to be part of the candidate key, as it is only “on the left side”. A alone is enough to “reach” all attributes, adding something therefore would violate minimality.

Question 3.2: Normal Form

What is the highest normal form of R ?

2NF (it is in 1NF and all FDs either depend on the whole key, or only on non-prime attributes). It is not in 3NF, as E depends on A transitively.

Question 3.3: Normalization

Normalize R into third normal form (3NF), if it is not already in 3NF.

$R'(\underline{A}, B, C, D) \quad R''(\underline{C}, \underline{D}, E)$

Question 4: Integrity Constraints

Question 4.1: Data Integrity

Name two options to enforce that a user can never insert a new row into *Shelter*, which has a city that does not exist in Denmark?

Three (of many options):

- 1) *Use an ENUM as domain for the city attribute to enumerate all valid cities*
- 2) *Use a trigger + signal*
- 3) *Create a new entity set City and use a foreign key on the city attribute in Shelter*

Question 4.2: Foreign Key Actions

Which referential action would you pick to ensure that a pet can not be deleted from the database, while it is still registered in a shelter (i.e., there is a row in the Shelter table that contains the pet's name)?

*Use **no** action as in*

FOREIGN KEY (petName) REFERENCES Pet(petName)

*or **RESTRICT** / **NO ACTION** (which is the default referential action) as in*

FOREIGN KEY (petName) REFERENCES Pet(petName) RESTRICT

FOREIGN KEY (petName) REFERENCES Pet(petName) NO ACTION

All options from above are equivalent.

Question 5: Indexing and Hashing

Assume that there is exactly one index on the Pet table (no automatic indexes were created), and the index was created via the following statement:

CREATE INDEX pet ON Pet(age) USING HASH;

Which of the following SQL queries is sped up by this index, i.e., it runs faster than before creating the index?

- a) SELECT * FROM Pet WHERE age <= 12;
-  b) SELECT * FROM Pet WHERE age = 12 AND breed = 'Pug';
- c) SELECT * FROM Pet WHERE age > 12 AND breed = 'Pug';
- d) SELECT * FROM Pet WHERE breed = 'Pug';
- e) SELECT * FROM Pet WHERE petName = 'George';
-  f) SELECT * FROM Pet WHERE age = 12;

Question 6: Formal Query Languages

Which of the following options in domain calculus is the equivalent of the relational algebra statement below?

$$\Pi_{petName}(\sigma_{city='Lyngby'}(reservedFor \bowtie Person))$$

- a) $\{petName \mid \langle petName, personName, reservedDate \rangle \in reservedFor \wedge \langle personName, 'Lyngby' \rangle \in Person\}$
-  b) $\{petName \mid \exists personName, reservedDate (\langle petName, personName, reservedDate \rangle \in reservedFor \wedge \langle personName, 'Lyngby' \rangle \in Person)\}$
- c) $\{petName \mid \exists personName, reservedDate (\langle petName, personName, reservedDate \rangle \in reservedFor) \wedge \exists personName (\langle personName, 'Lyngby' \rangle \in Person)\}$
- d) $\{petName \mid \exists personName, reservedDate, city (\langle petName, personName, reservedDate \rangle \in reservedFor \wedge \langle personName, city \rangle \in Person)\}$